Executive overview

INTERNAL AI APPLICATION

PRODUCTION

Settlement Failure Early Warning

AI-INT-2026-041 | Model v2.3.1

0.95	0.91	Medium	Daily
ROC-AUC	Precision	Residual risk	Metric checks

BUSINESS OBJECTIVE

Identify likely T+1 settlement exceptions early enough for operations analysts to prioritize investigations. Outputs are advisory and cannot release, block, or alter settlement instructions without a human decision.

DECISION WORKFLOW

01	02	03	04
Transaction / attributes	Risk score / & alert	Analyst / review	Manual / action

LATEST OVERSIGHT FINDINGS

WATCH ITEM

A September batch-feed delay triggered manual fallback. Incident closed; drift PSI is 0.14 against a review threshold of 0.20.

Structured inventory | ownership, data and controls

CANONICAL INVENTORY FIELDS

22 mapped fields • values are fictional • explicit IDs support extraction testing

01 / Submission ID
AI-INT-2026-041

02 / Legal entity
Example Post-Trade Operations (fictional)

03 / Application name
Settlement Failure Early Warning (SFEW)

04 / Business use case
T+1 settlement exception prediction

05 / Business owner
Maya Chen, Director, Post-Trade Analytics

06 / Technical owner
Alex Rivera, Senior ML Engineer

07 / Model type
Gradient-boosted trees (XGBoost)

08 / AI provider
Internally developed; no external inference API

09 / Model version
2.3.1

10 / Lifecycle status
Production

11 / First deployment
2026-07-15

12 / Input data
Transaction metadata + exception history

13 / Data sensitivity
Confidential, transaction-level data

14 / Financial criticality
High; advisory operations prioritization

15 / Human oversight
Analyst approves every escalation

16 / Validation status
Passed independent validation on 2026-10-20

17 / Performance
Precision 0.91; recall 0.86; ROC-AUC 0.95

18 / Risk rating
Inherent HIGH; residual MEDIUM

19 / Incident history
1 closed incident; see page 4

20 / Monitoring frequency
Daily metrics; monthly drift review

21 / Fallback process
Manual rules-based exception queue

22 / Next periodic review
2027-01-20

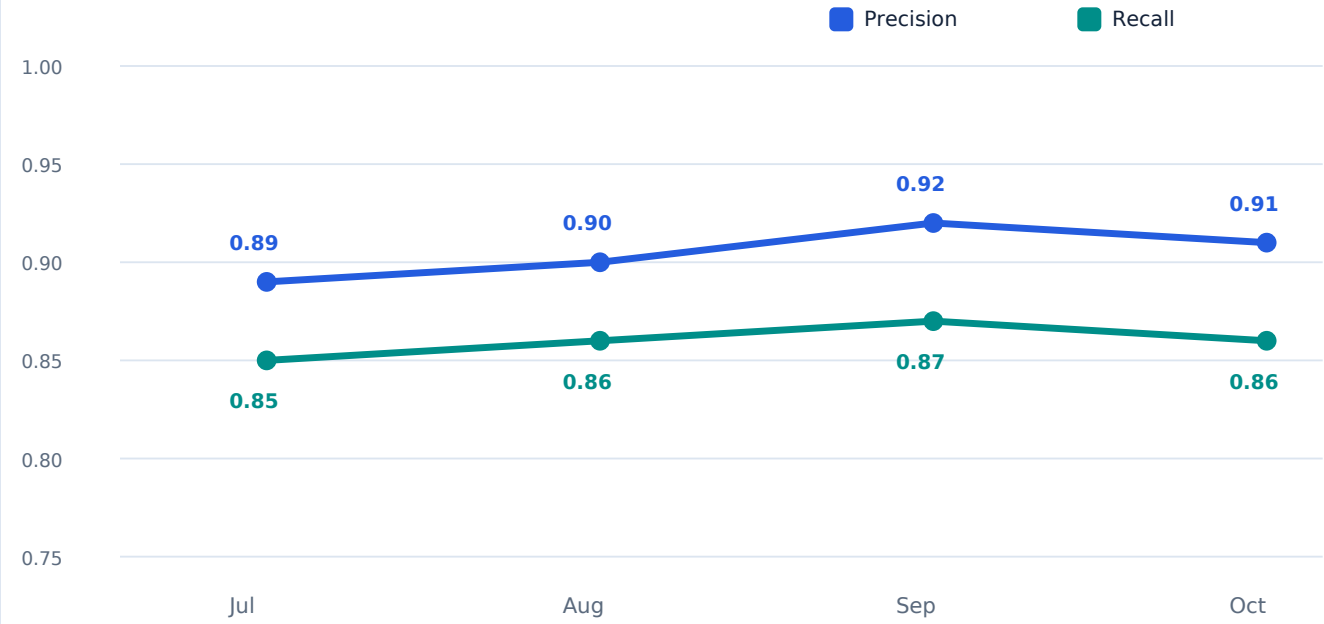
DATA-HANDLING NOTE

Synthetic data only in this sample. Real transaction-level inputs would require approved access controls, retention limits, and audit logging.

Model validation | metrics and drift

MONTHLY MODEL PERFORMANCE

Holdout/production shadow sample; monthly figures are synthetic.



VALIDATION RESULTS

Measure	Observed	Threshold	Result
Precision	0.91	>= 0.85	**Pass**
Recall	0.86	>= 0.80	**Pass**
ROC-AUC	0.95	>= 0.90	**Pass**
Feature drift (PSI)	0.14	Review >= 0.20	**Below trigger**

VALIDATION EVIDENCE

Independent review completed 2026-10-20 using a 30,000-record retrospective test set. No material threshold breach; next review scheduled for 2027-01-20.

Governance | risk register and incident review

MATERIAL RISK & CONTROL REGISTER

Risk	Mitigation / control	Residual	Control owner
Model drift	Monthly PSI + outcome monitoring	Medium	Model Risk
Data leakage	RBAC, encryption, access audit	Low	InfoSec
False alerts	Human review before escalation	Medium	Operations
Operational resilience	Manual queue + batch feed checks	Medium	Platform Ops

INCIDENT REGISTER

INC-2026-09-12-01 • Closed 2026-09-13

Category: Data pipeline / batch feed latency

Impact: 42-minute delay in risk-score availability; 0 settlement actions affected.

Response: Manual exception queue activated; delayed alerts were reprocessed.

Preventive action: Feed freshness alarm reduced from 30 to 10 minutes.

MONITORING & OVERSIGHT CADENCE

Every business day	Precision/recall proxies, alert volumes, feed freshness	Operations + ML Ops
Monthly	PSI drift review, false-positive sampling, access logs	Model Risk
Quarterly	Control attestation and material change review	AI Governance

Decision record: Residual MEDIUM risk accepted for advisory use only; human approval is mandatory. Model changes require revalidation before release.